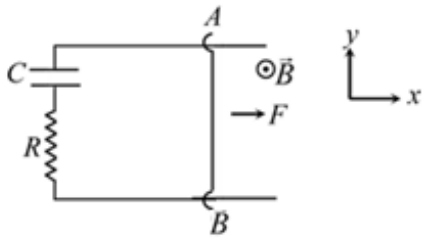
# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

A conducting rod $AB$ moves parallel to $x$ -axis in the $x-y$ plane. $A$ uniform magnetic field $B$ pointing normally out of the plane exists throughout the region. A force $F$ acts perpendicular to the rod, so that the rod moves with uniform velocity $v$. The force $F$ is given by
(neglect resistance of all the wires)




**(A)** $\frac{vB^2l^2}{R}e^{-t/RC}$

**(B)** $\frac{vB^2l^2}{R}$

**(C)** $\frac{vB^2l^2}{R}\left(1-e^{-t/RC}\right)$

**(D)** $\frac{vB^2l^2}{R}\left(1-e^{-2t/RC}\right)$

**Q.2**

A metal rod of length l cuts across a uniform magnetic field $B$ with a velocity $v$. If the resistance of the circuit of which the rod forms a part is $\mathrm{r}$, then the force required to move the rod is

**(A)** $\frac{\mathrm{B}^2l^2v}{r}$

**(B)** $\frac{\mathrm{B}lv}{r}$

**(C)** $\frac{\mathrm{B}^2lv}{r}$

**(D)** $\frac{\mathrm{B}^2l^2v^2}{r}$

**Q.3**

A rectangular, a square, a circular and an elliptical loop, all in the $(\mathrm{x}-\mathrm{y})$ plane, are moving out of a uniform magnetic field with a constant velocity, $\overrightarrow{\mathrm{V}}=\mathrm{v}\hat{\mathrm{i}}$. The magnetic field is directed along the negative $z$ axis direction. The induced emf, during the passage of these loops, out of the field region, will not remain constant for

**(A)** the circular and the elliptical loops.

**(B)** only the elliptical loop.

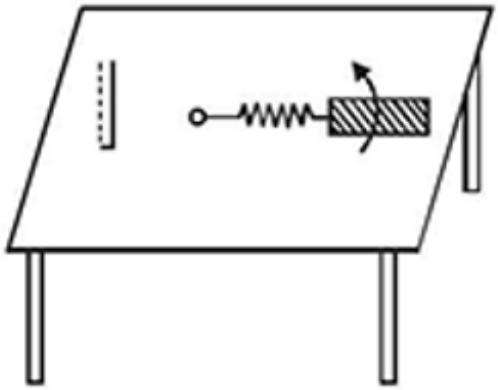
**(C)** any of the four loops.

@IITJEE_Advanced

**(D)** the rectangular, circular and elliptical loops.

**Q.4**

A metallic rod of length ' $\ell$ ' is tied to a string of length $2\ell$ and made to rotate with angular speed $\omega$ on a horizontal table with one end of the string fixed. If there is a vertical magnetic field ' $B$ ' in the region, the e.m.f. induced across the ends of the rod is



**(A)** $\frac{2B\omega\ell^2}{2}$

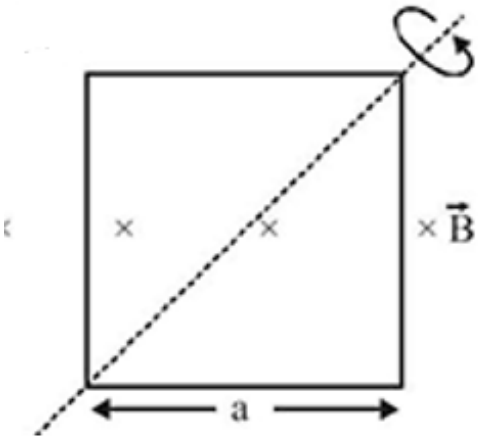
**(B)** $\frac{3B\omega\ell^2}{2}$

**(C)** $\frac{4B\omega\ell^2}{2}$

**(D)** $\frac{5B\omega\ell^2}{2}$

**Q.5**

A square loop of side a is rotating about its diagonal with angular velocity $\omega$ in a perpendicular magnetic field $\overrightarrow{\mathrm{B}}$. It has 10 turns. The emf induced is




**(A)** $\mathrm{Ba}^2 \sin \omega t$

**(B)** $\mathrm{Ba}^2 \cos \omega t$

**(C)** $5\sqrt{2}\,\mathrm{Ba}^2$

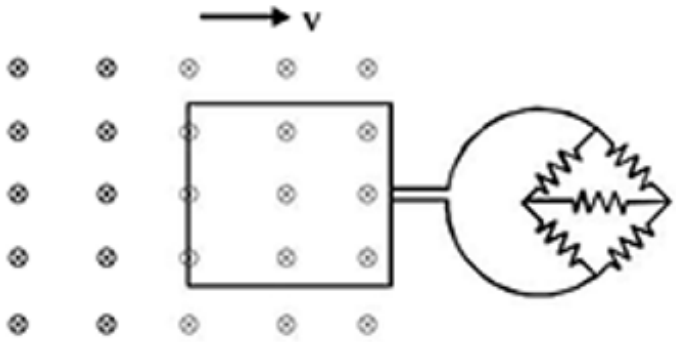
**(D)** $10\,\mathrm{Ba}^2\omega \sin \omega t$

**Q.6**

A square metal loop of side $10\ \mathrm{cm}$ and resistance $1\Omega$ is moved with a constant velocity partly inside a uniform

@IITJEE_Advanced

magnetic field of $2\mathrm{Wbm}^{-2}$, directed into the paper, as shown in the figure. The loop is connected to a network of five resistors each of value $3\Omega$. If a steady current of $1\ \mathrm{mA}$ flows in the loop, then the speed of the loop is




**(A)** $0.5\mathrm{cms}^{-1}$

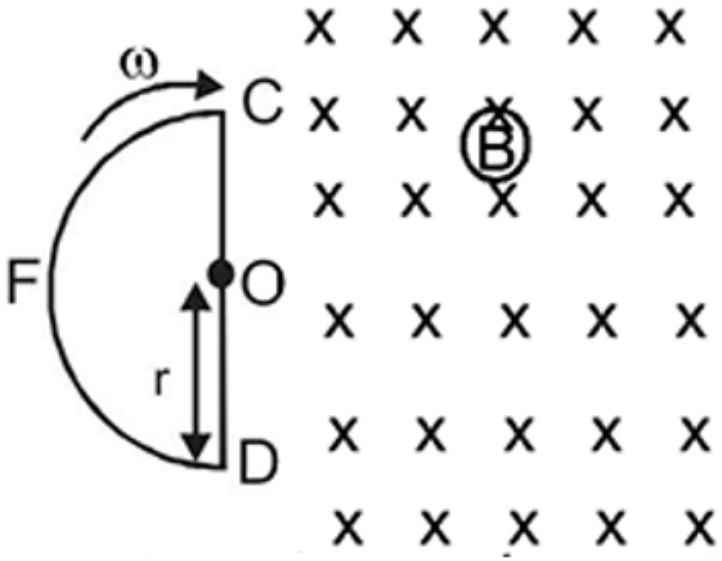
**(B)** $1\mathrm{cms}^{-1}$

**(C)** $2\mathrm{cms}^{-1}$

**(D)** $4\mathrm{cms}^{-1}$

**Q.7**

In fig, CODF is a semicircular loop of a conducting wire of resistance $\mathrm{R}$ and radius $\mathrm{r}$. It is placed in a uniform magnetic field $\mathrm{B}$, which is directed into the page (perpendicular to the plane of the loop).




The loop is rotated with a constant angular speed $\omega$ about an axis passing through the centre $\mathrm{O}$, and perpendicular to the page. Then the induced current in the wire loop is

**(A)** zero

**(B)** $\mathrm{Br}^2\omega/\mathrm{R}$

**(C)** $\mathrm{Br}^2\omega/2\mathrm{R}$

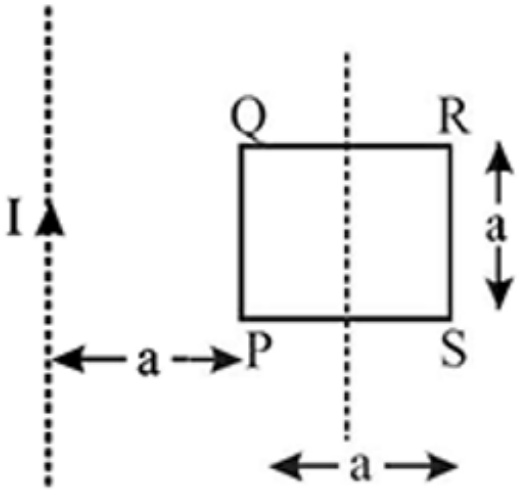
**(D)** $\mathrm{B}\pi\mathrm{r}^2\omega/\mathrm{R}$

**Q.8**

In the figure shown a square loop PQRS of side 'a' and resistance '1 unit' is placed near an infinitely long wire.

@IITJEE_Advanced

carrying a constant current I. The sides PQ and RS are parallel to the wire. The wire and the loop are in the same plane. The loop is rotated by $180^\circ$ about an axis parallel to the long wire and passing through the mid points of the side $QR$ and $PS$. The total amount of charge which passes through any point of the loop during rotation is




**(A)** $\frac{\mu_0 \mathrm{Ia}}{2\pi} \ln 2$

**(B)** $\frac{\mu_0 \mathrm{Ia}}{\pi} \ln 2$

**(C)** $\frac{\mu_0 \mathrm{Ia}^2}{2\pi}$

**(D)** cannot be found because time of rotation not given

**Q.9**

A boat is moving due east in a region where the earth's magnetic field is $5.0 \times 10^{-5} \mathrm{NA}^{-1}\ \mathrm{m}^{-1}$ due north and horizontal. The boat carries a vertical aerial $2\ \mathrm{m}$ long. If the speed of the boat is $1.50\ \mathrm{ms}^{-1}$. Find the magnitude of the induced emf(in $\mathrm{mV}$ ) in the wire of aerial.

**(A)** 0.15

**(B)** 1.5

**(C)** 0.015

**(D)** 15

**Q.10**

A rod $PQ$ of length $L$ moves with a uniform velocity $v$ parallel to a long straight wire carrying a current $i$, the end $P$ remaining at a distance r from the wire. The emf induced across the rod is

**(A)** $\frac{\mu_0 i v^2}{2\pi} \ln\left(\frac{r+L}{r}\right)$

**(B)** $\frac{\mu_0 i^2 v^2}{2\pi} \ln\left(\frac{r^2+L}{r}\right)$

**(C)** $\frac{\mu_0 i v}{2\pi} \ln\left(\frac{r+L}{r}\right)$

**(D)** $\frac{\mu_0 i v}{2\pi} \ln\left(\frac{r^2+L^2}{L^2}\right)$

@IITJEE_Advanced

**Q.11**

A conductor $\mathrm{AB}$ of length $l$ moves in $\mathrm{x}-\mathrm{y}$ plane with velocity $\overrightarrow{\mathrm{v}}=\mathrm{v}_0(\hat{\mathrm{i}}-\hat{\mathrm{j}})$. A magnetic field $\overrightarrow{\mathrm{B}}=\mathrm{B}_0(\hat{\mathrm{i}}+\hat{\mathrm{j}})$ exists in the region. The induced emf is

**(A)** zero

**(B)** $\mathrm{B}_0 l \mathrm{v}_0$

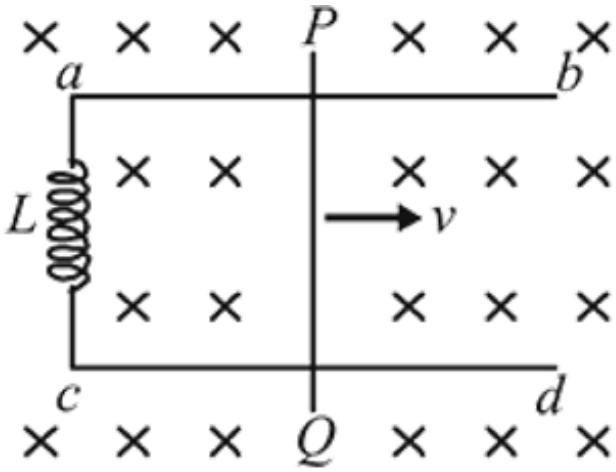
**(C)** $\mathrm{B}_0 l \mathrm{v}_0$

**(D)** $\sqrt{2}\,\mathrm{B}_0 \mathrm{l} \mathrm{v}_0$

**Q.12**

In a uniform and constant magnetic field of induction $B$, two long conducting wires $ab$ and $cd$ are kept parallel to each other at distance $\ell$ with their plane perpendicular to $B$. The ends $a$ and $c$ are connected together by an ideal inductor of inductance $L$. A conducting slider wire $PQ$ is imparted a speed $v_0$ at time $t=0$. The situation is shown in the figure.

At time $t=\frac{\pi\sqrt{mL}}{4B\ell}$, the value of current $I$ through the wire $PQ$ is (ignore any resistance, electrical as well as mechanical)




**(A)** $\sqrt{\frac{mv_0^2}{L}}$

**(B)** $\sqrt{\frac{mv_0^2}{2L}}$

**(C)** $\sqrt{\frac{mv_0^2}{4L}}$

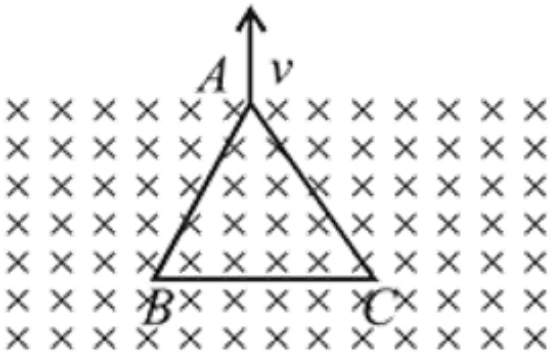
**(D)** zero.

**Q.13**

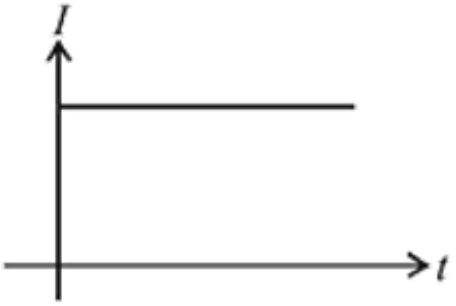
An equilateral triangular loop $ABC$ made of uniform thin wires is being pulled out of a region with a uniform speed $v$, where a uniform magnetic field $\vec{B}$ perpendicular to the plane of the loop exists. At time $t=0$, the point

@IITJEE_Advanced

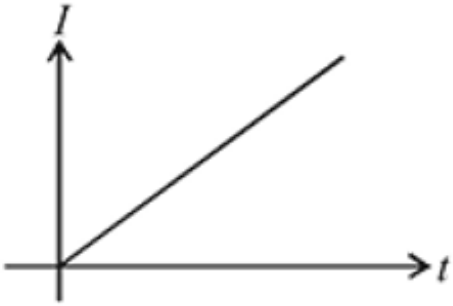
$A$ is at the edge of the magnetic field. The induced current (I) vs time $(t)$ graph will be as



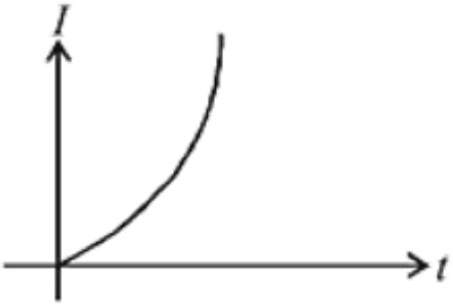

**(A)**




**(B)**




**(C)**




**(D)**




**Q.14**

A vertical ring of radius r and resistance R falls vertically. It is in contact with two vertical rails which are joined at the top. The rails are without friction and resistance. There is a horizontal uniform magnetic field of magnitude B perpendicular to the plane of the ring and the rails. When the speed of the ring is v, the current in the top horizontal of the rail section is

**(A)** 0

@IITJEE_Advanced

**(B)** $\frac{2Brv}{R}$

**(C)** $\frac{4Brv}{R}$

**(D)** $\frac{8Brv}{R}$

**Q.15**

Charge $Q$ is uniformly distributed on a thin insulating ring of mass $m$ which is initially at rest. To what angular velocity will the ring be accelerated when a magnetic field B, perpendicular to the plane of the ring, is switched on?

**(A)** $\frac{QB}{2\,m}$

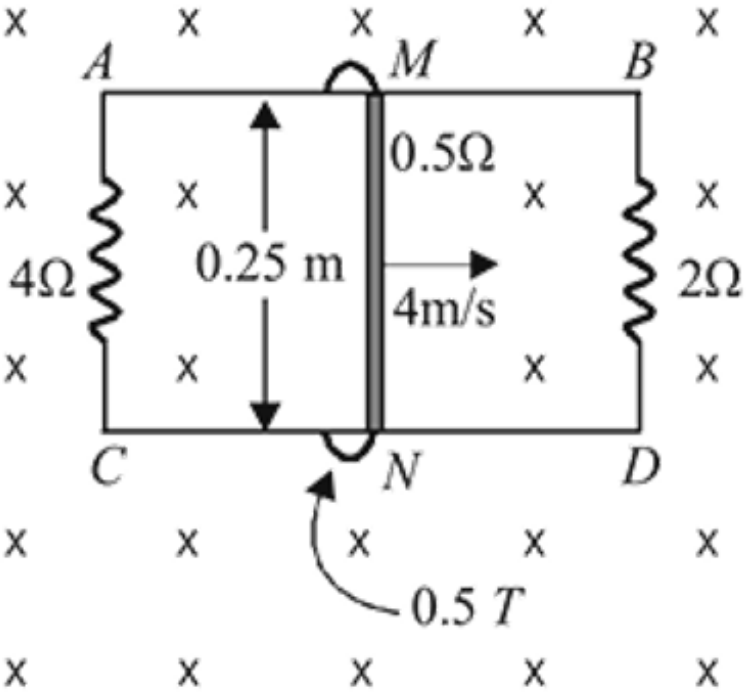
**(B)** $\frac{3QB}{2m}$

**(C)** $\frac{QB}{m}$

**(D)** $\frac{QB}{4\,m}$

**Q.16**

A sliding wire of length $0.25\ m$ and having a resistance of $0.5\Omega$ moves along conducting guiding rails $AB$ and $CD$ with a uniform speed of $4\ m/s$. A magnetic field of $0.5\ T$ exists normal to the plane of $ABCD$ directed into the page. The guides are short -circuited with resistances of 4 ohm and $2\Omega$ as shown. The current through the sliding wire is :




**(A)** $0.27\ A$

**(B)** $0.37\ A$

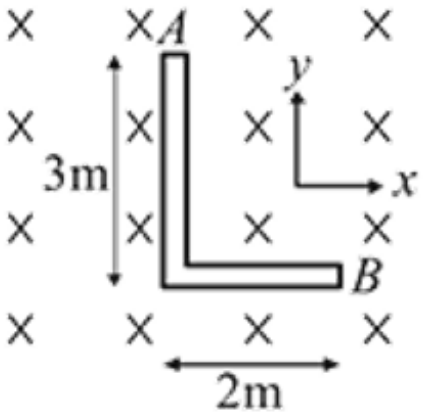
**(C)** $1.0\ A$

**(D)** $0.72\ A$

**Q.17**

@IITJEE_Advanced

An $L$ -shaped conductor rod is moving in transverse magnetic field as shown in the figure. Potential difference between ends of the rod is maximum if the rod is moving with velocity




**(A)** $4\hat{i} - 6\hat{j}\ \mathrm{m/s}$

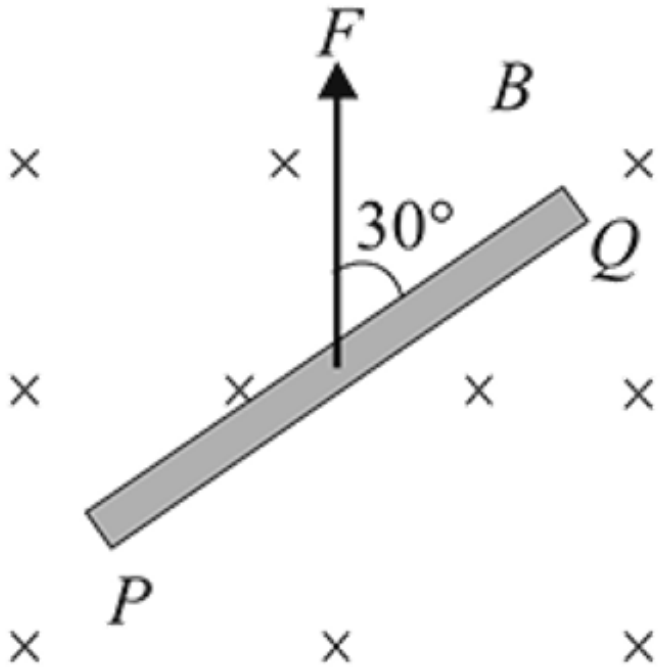
**(B)** $-4\hat{i} + 6\hat{j}\ \mathrm{m/s}$

**(C)** $3\hat{i} + 2\hat{j}\ \mathrm{m/s}$

**(D)** $\sqrt{13}\hat{i}\ \mathrm{m/s}$

**Q.18**

A conducting rod of mass $m$ and $\ell$ is placed on $a$ smooth horizontal surface in a region where transverse uniform magnetic field $B$ exists in the region. At $t = 0$, constant force $F$ starts acting on the rod at its mid point as shown. Potential difference between ends of the rod, $V_P - V_Q$ at any time $t$ is




**(A)** $\frac{BF\ell t}{2m}$

**(B)** $\frac{BF\ell t}{4m}$

**(C)** $\frac{5BF\ell t}{8m}$

@IITJEE_Advanced

**(D)** $\frac{7BFlt}{8m}$

**Q.19**

A $0.1$ m long conductor carrying a current of 50 A is perpendicular to a magnetic field of $1.21\mathrm{mT}$. The mechanical power to move the conductor with a speed of $1\ \mathrm{ms}^{-1}$ is

**(A)** $0.25\ \mathrm{mW}$

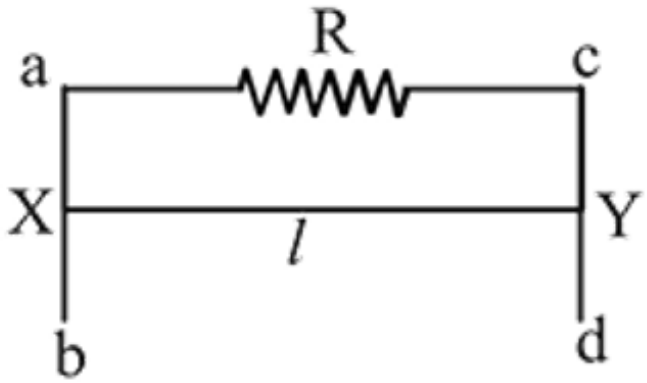
**(B)** $6.25\ \mathrm{mW}$

**(C)** $0.625\ \mathrm{W}$

**(D)** $1\ \mathrm{W}$

**Q.20**

A conducting wire $xy$ of length $l$ and mass $m$ is sliding without friction on vertical conduction rails ab and cd shown in Fig. Auniform magnetic field $B$ exists perpendicular to the plane of the rails, $x$ moves with a constant velocity of




**(A)** $\frac{mgR}{Bl}$

**(B)** $\frac{mgR}{Bl^2}$

**(C)** $\frac{mgR}{B^2l^2}$

**(D)** $\frac{mgR}{B^2l}$

## Answers & Solutions

**Q.1 Answer:**

$\frac{vB^2l^2}{R}e^{-t/RC}$

**Solution:**

**Induced emf in the rod** $\varepsilon = Blv$

**Current in the circuit** $I = \frac{\varepsilon}{R}e^{-t/RC} = \frac{Blv}{R}e^{-t/RC}$

**Since the net force on the rod should be zero, the external force will be equal in magnitude but opposite to the magnetic force.**

$F = IlB = \frac{B^2l^2v}{R}e^{-t/RC}$

**Q.2 Answer:**

@IITJEE_Advanced

$\frac{B^2 l^2 v}{r}$

**Solution:**

**Indueed emt** $= B \vee l$

$$i = \frac{Bv_l}{r}$$

$$f_m = i\ell B$$

$$= \frac{Bve \times l \times B}{r}$$

$$= \frac{B^2 e^2 V}{r}$$

**Q.3 Answer:**

the circular and the elliptical loops.

**Solution:**

**The induced emf will remain constant only in the case of rectangular and square loops. In case of the circular and the elliptical loops, the rate of change of area of the loops during their passage out of the field is not constant, hence induced emf will not remain constant for them.**

**Q.4 Answer:**

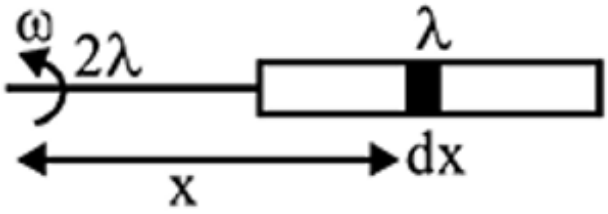
$\frac{5B\omega\ell^2}{2}$

**Solution:**

**Here, induced e.m.f.**

$$e = \int_{2\ell}^{3\ell} (\omega x) B dx$$

$$= B\omega \frac{\left[(3\ell)^2 - (2\ell)^2\right]}{2} = \frac{5B\ell^2\omega}{2}$$




**Q.5 Answer:**

$10\ \mathrm{Ba}^2\omega \sin \omega \mathrm{t}$

**Solution:**

$$\phi = \mathrm{nBA}\cos\theta = 10\ \mathrm{Ba}^2 \cos\omega \mathrm{t}$$

$$\mathrm{e} = -\frac{\mathrm{d}\phi}{\mathrm{dt}} = -\frac{\mathrm{d}}{\mathrm{dt}}\left(10\mathrm{Ba}^2 \cos\omega \mathrm{t}\right) = 10\ \mathrm{Ba}^2 \sin\omega \mathrm{t}(\omega)$$

**Q.6 Answer:**

@IITJEE_Advanced

$2 \mathrm{cms}^{-1}$

**Solution:**

$\varepsilon = \mathrm{B}\ell\mathrm{v} = 2 \times 10^{-1} \times \mathrm{v} = 0.2\mathrm{v}$

$I = \frac{\varepsilon}{R} = 10^{-3} \Rightarrow \frac{0.2v}{4} = 10^{-3}$

**[Since effective resistance** $\mathrm{R}$ **of bridge is**

$\mathrm{R} = \frac{6 \times 6}{6+6} = 3\Omega$

**so total resistance** $= 1 + 3 = 4\Omega$ **]** $\Rightarrow \mathrm{v} = 2 \text{ cm s}^{-1}$

**Q.7 Answer:**

$\mathrm{Br}^2\omega/2\mathrm{R}$

**Solution:**

**The area swept by radius** $\mathrm{OC}$ **in one half circle is** $\pi \mathrm{r}^2/2$**. The flux change in time** $\mathrm{T}/2$ **is thus** $\left(\pi \mathrm{r}^2\ \mathrm{B}/2\right)$**. The induced emf is then** $\mathrm{e} = \pi \mathrm{r}^2\ \mathrm{B}/\mathrm{T} = \mathrm{B}\omega \mathrm{r}^2/2 \left[\because T = \frac{2\pi}{\omega}\right]$

**The induced current is then** $\mathrm{I} = \mathrm{e}/\mathrm{R} = \mathrm{B}\omega \mathrm{r}^2/2\mathrm{R}$

**Q.8 Answer:**

$\frac{\mu_0 \mathrm{Ia}}{\pi} \ln 2$

**Solution:**

$\mathrm{dq} = \frac{-\mathrm{d}\phi}{\mathrm{R}} \quad \left[\because \phi_\mathrm{f} = -\phi_\mathrm{i}\right]$

$\therefore \mathrm{dq} = \frac{2\phi_\mathrm{i}}{\mathrm{R}} = \frac{2\mu_0 \mathrm{Ia}}{\mathrm{R}.2\pi} \ln\left(\frac{2\mathrm{a}}{\mathrm{a}}\right) = \frac{\mu_0 \mathrm{Ia} \cdot \ln 2}{\mathrm{R}\pi}$

$\left[\phi = \int_\mathrm{a}^{2\mathrm{a}} \frac{\mu_0 \mathrm{I}}{2\pi \mathrm{x}} \mathrm{adx} = \frac{\mu_0 \mathrm{Ia}}{2\pi} \ln \frac{2\mathrm{a}}{\mathrm{a}}\right]$

**Q.9 Answer:**

0.15

**Solution:**

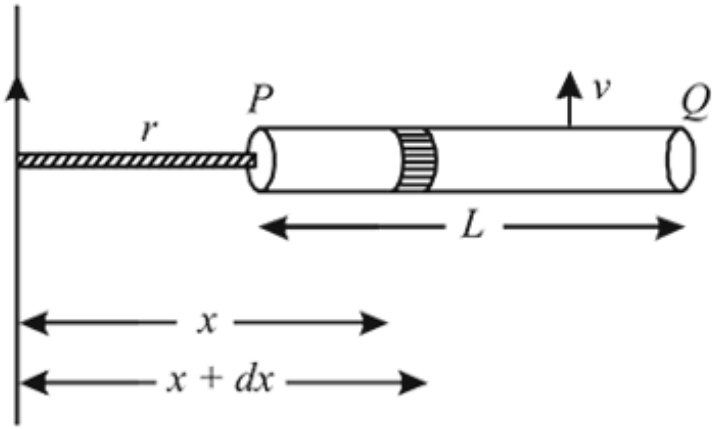
$$\begin{aligned} \text{Induced emf } &= vB_H l = 1.5 \times 5 \times 10^{-5} \times 2 \\ &= 15 \times 10^{-5} \\ &= 0.15\mathrm{mV} \end{aligned}$$

**Q.10 Answer:**

$\frac{\mu_0 iv}{2\pi} \ln\left(\frac{r+L}{r}\right)$

**Solution:**

@IITJEE_Advanced




**Consider a small element of length $dx$ of the rod at a distance $x$ and $(x+dx)$ from the wire. The emf induced across the element**

$de = Bvdx$ **...(i)**

**We know that magnetic field $B$ at a distance $x$ from a wire carrying a current $i$ is given by**

$$B = \frac{\mu_0}{2\pi} \cdot \frac{i}{x} ...(ii)$$

**From eqs. (i) and (ii),**

$$de = \frac{\mu_0 i}{2\pi x} vdx ...(iii)$$

**The emf $e$ induced in the entire length of the rod $PQ$ is given by** $e = \int de = \int_P^Q \frac{\mu_0}{2\pi}\frac{i}{x}vdx$
$= \int_r^{r+L} \frac{\mu_0}{2\pi}\frac{i}{x}vdx = \frac{\mu_0}{2\pi}iv\int_r^{r+L}\frac{dx}{x}$
$= \frac{\mu_0 iv}{2\pi}\log_e\left(\frac{r+L}{r}\right)$

**Q.11 Answer:**

zero

**Solution:**

$\vec{\ell}$, $\overrightarrow{\mathrm{v}}$, **and** $\overrightarrow{\mathrm{B}}$ **are coplanar.**

**Q.12 Answer:**

$\sqrt{\frac{mv_0^2}{2L}}$

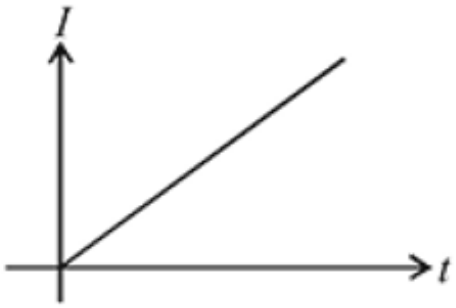
**Solution:**

$$I(t) = \sqrt{\frac{m}{L}} v_0 \sin\omega t \text{ where } \omega = \frac{B\ell}{\sqrt{mL}}$$

$$\Rightarrow I = \sqrt{\frac{mv_0^2}{2L}}$$

@IITJEE_Advanced

**Q.13 Answer:**

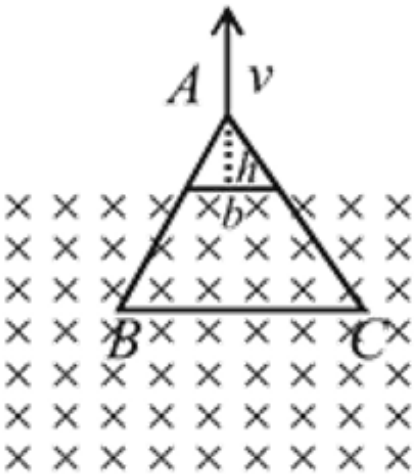



**Solution:**

$I = \frac{\varepsilon}{R} = \frac{|d\phi/dt|}{R} = \frac{B \cdot \frac{dA}{dt}}{R}$

$= \frac{B\frac{d}{dt}\left(\frac{1}{2}h.b\right)}{R} \propto \frac{Bb\frac{dh}{dt}}{R} \propto Bbv$

$\therefore b \propto t \Rightarrow I \propto t$

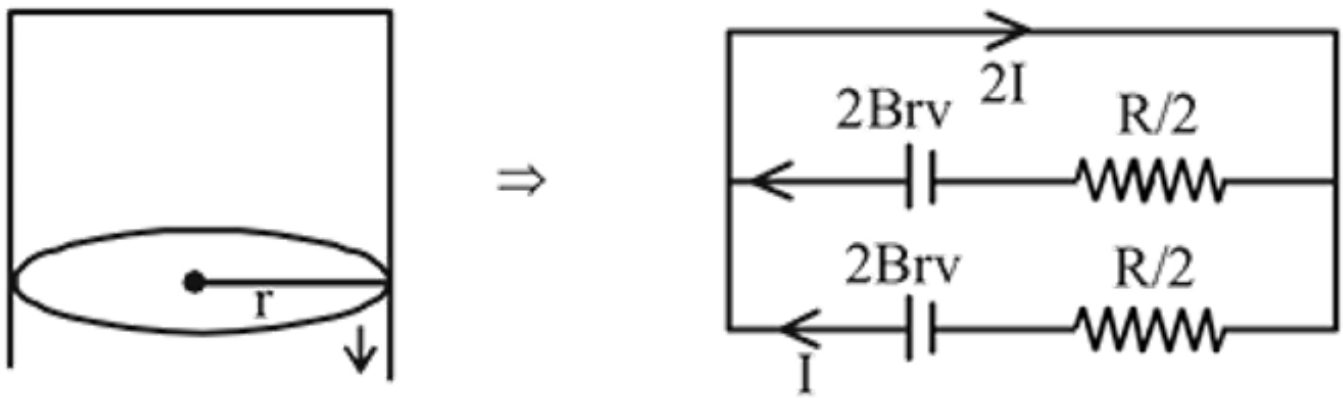



**Q.14 Answer:**

$\frac{8Brv}{R}$

**Solution:**

$I = \frac{2Brv}{R/2} = \frac{4Brv}{R}$




**Current in the top horizontal** $= 2I = \frac{8Brv}{R}$

**Q.15 Answer:**

$\frac{QB}{2\,m}$

@IITJEE_Advanced

**Solution:**

**Let $r$ be the radius of ring.**

$\int \mathrm{E.dl} = \mathrm{E} \times 2 \times \pi \mathrm{r}$

$\int \frac{\mathrm{B}}{\mathrm{dt}} \times \mathrm{dA} = \left|\frac{\mathrm{B}}{\mathrm{t}}\right| \times \pi \mathrm{r}^2$

**Equating $E$ and $B$ :**

$\mathrm{E} = \frac{\mathrm{dB}}{\mathrm{dt}} \times \mathrm{r/L}$

**Now,** $Q \times E = m \times r \times a$

**where** $\mathrm{m}$ **is mass of ring and a is angular acceleration.**

**So,** $a = \left(Q \times \frac{dB}{dt}\right) / 2m$

**Let angular velocity be w, so**

$$\mathrm{w} = \int \mathrm{a.dt}$$

$$\Rightarrow \mathrm{w} = \mathrm{Q/2\ m} \times \int (\mathrm{dB/dt}) \cdot \mathrm{dt}$$

$$\Rightarrow \mathrm{w} = \mathrm{QB/2\ m}$$

**Q.16 Answer:**

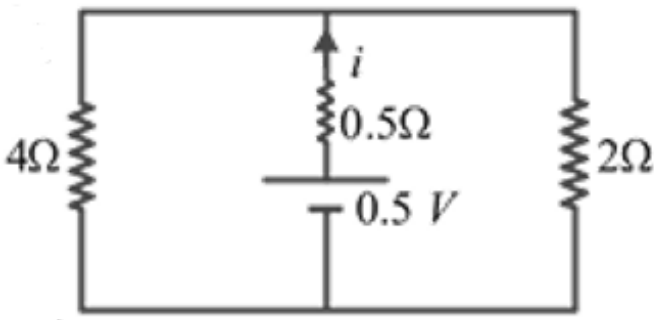
$0.27\ \mathrm{A}$

**Solution:**

**The induced emf across the sliding wire** $e = Bv\ell = 0.5 \times 4 \times 0.25 = 0.5\ \mathrm{V}$

**The effective circuit shown in figure. The equivalent resistance of the circuit** $r = \frac{4\times2}{4+2} + 0.5 = 1.83\Omega$

**Now,** $i = \frac{V}{R} = \frac{0.5}{183} = 0.27\ \mathrm{A}$




**Q.17 Answer:**

$3\hat{i} + 2\hat{j}\ \mathrm{m/s}$

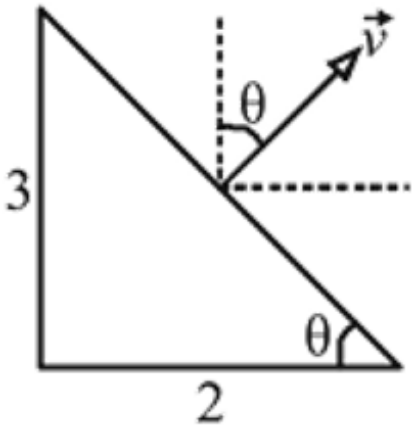
**Solution:**

**For maximum potential difference the velocity must be perpendicular to the line** $AB$**, and so**

$\vec{v} = v \sin\theta\hat{i} + v\cos\theta\hat{j}$

$$= v\left[\frac{3}{\ell}\hat{i} + \frac{2}{\ell}\hat{j}\right] = \frac{v}{\ell}[3\hat{i} + 2\hat{j}]$$

@IITJEE_Advanced




**Q.18 Answer:**

$\frac{BF\ell t}{2m}$

**Solution:**

$$v = at = \frac{F}{m}t$$

**Velocity at any time** $t$, Now, $e = V_P - V_Q = Bv\ell \sin 30^\circ = B\left(\frac{Ft}{m}\right)\ell \times \frac{1}{2}$

$$= \frac{BF\ell t}{2m}$$

**Q.19 Answer:**

$6.25\ \text{mW}$

**Solution:**

**Mechanical power required = electrical power expended due to the induced** emf

$$= (Blv) \times I$$

$$=(0.00121 \times 0.1 \times 1) \times 50$$

$$= 6.25\ \text{mW}$$

**Q.20 Answer:**

$\frac{mgR}{B^2l^2}$

**Solution:**

$\text{BI}\ell = \text{mg}$ **or** $\text{B}\frac{\text{Bv}\ell}{\text{R}}\ell = \text{mg}$ **or** $\text{v} = \frac{\text{mgR}}{\text{B}^2\ell^2}$

@IITJEE_Advanced